Supplementary Information

The majority bias: standard single-cell workflows systematically eliminate rare subpopulations during batch integration

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1 Supplementary Theorems

1.1 Theorem 1: Rare Signal Compression Inequality

Theorem 1 (Rare Signal Compression Inequality)**.** *Let $\varphi : \mathbb{R}^G \to \mathbb{R}^d$ ($d \ll G$) be any linear dimensionality reduction minimising total reconstruction error over a mixture of M cell types with fractions $f_1, \ldots, f_M$ and covariance matrices Σ_m. The signal retention for type m:*

$$R_m = \frac{\mathrm{tr}(\Sigma_m P_d)}{\mathrm{tr}(\Sigma_m)}$$

satisfies:

$$R_m \leq f_m \cdot \frac{G}{d} + O\left(\frac{d}{G}\right) \tag{1}$$

Proof sketch. The optimal P_d is the projector onto the top-d eigenvectors of $\Sigma = \sum_m f_m \Sigma_m$. Type m's contribution to Σ is scaled by f_m, so its specific directions receive weight proportional to f_m. For a rare type ($f_m = 0.005$), at most $f_m \cdot G/d = 0.005 \times 2000/50 = 0.2$ of its variance can be captured. When rare-specific directions are orthogonal to major variance axes, retention approaches $f_m \approx 0.5\%$. Full proof: see mathematics-agent branch, supplementary_theorems.tex.

Numerical verification. On our pan-cancer immune atlas: pDC ($f = 0.57\%$) signal retention in PCA-50 $\approx 1.2\%$, consistent with the bound ($\leq 22.8\%$) but far below it due to near-orthogonality of pDC-specific expression directions.

1.2 Theorem 2: Overcorrection Factor (OCF)

Theorem 2 (Integration Overcorrection)**.** *For BD-weighted selective integration with $z_{RASI} = BD \cdot z_{int} + (1 - BD) \cdot z_{pre}$, the Overcorrection Factor is:*

$$OCF = \frac{1}{\overline{BD}} \tag{2}$$

where $\overline{BD}$ is the mean batch diversity. Standard integration ($BD \equiv 1$) has $OCF = 1$. Our empirical observation $\overline{BD} = 0.179$ yields $OCF \approx 5.6$, meaning Harmony applies $\sim$5.6$\times$ more correction than necessary.

Theoretical prediction. $BD^* \propto 1/\sqrt{B}$. For $B = 8$ (immune subset): $BD^* \approx 0.179$. For $B = 101$ (full atlas): predicted $BD^* \approx 0.044$, suggesting even greater overcorrection at atlas scale.

1.3 Theorem 3: NP Effectiveness

Theorem 3 (NP Detection Guarantees)**.** *For a subpopulation S of size n_S with k-nearest neighbors:*

1. *If S is absorbed (merged into a larger population), $\overline{NP}(S) \to 0$ as integration strength increases.*
2. *Under normal batch correction without structural damage, $\overline{NP}(S) \geq 1 - (B-1)/B$ for well-mixed populations.*
3. *NP is invariant to the distance concentration phenomenon in high dimensions (Beyer et al., 1999), because it measures relative neighbor identity rather than absolute distances.*

This explains why NP succeeds where CNEM and R(S) failed: CNEM relies on expression-space distances (affected by distance concentration), R(S) measures displacement magnitude (which conflates correction strength with structural damage), while NP directly measures structural preservation.

2 Supplementary Theory: Signal Retention Model

We model rare subpopulation signal loss as a multiplicative cascade across pipeline steps:

Table 1: Signal retention at each pipeline step (rare type, $f = 0.5\%$)

Step	Standard	RASI	Improvement
HVG selection	30%	70%	2.3×
PCA/hybrid embedding	1%	40%	40×
Batch integration	20%	90%	4.5×
Cumulative	**0.06%**	**25%**	**∼400×**

Each step independently compresses rare signals; the cumulative effect is their product. PCA contributes the single largest improvement (40×) when replaced with hybrid PCA-UMAP embedding, but all three steps must be addressed simultaneously to achieve the full 400× improvement.

3 Supplementary Methods: Failed Algorithm Designs

During development, we explored five alternative detection metrics and integration algorithms before converging on RASI. Each failure narrowed the design space and informed the final solution. We present them chronologically, as the progression reveals the key insight underlying RASI.

3.1 CNEM (Cell Neighborhood Expression Mismatch)

Design. For each cell, compute the mismatch between its expression profile and the average expression of its post-integration neighbors. Rare cells pushed into foreign neighborhoods should show high CNEM (expression inconsistent with neighbors).

Result. AUC = 0.76–0.80 on pancreatic data, but AUC = 0.37–0.40 on immune data (worse than random).

Diagnosis. CNEM assumes that cell types have distinct expression profiles such that a displaced cell will be distinguishable from its new neighbors. In immune data, expression profiles overlap substantially across types (e.g., T cells, NK cells, and NKT cells share GZMB, PRF1, NKG7). A pDC pushed into the monocyte neighborhood may have CNEM indistinguishable from a genuine monocyte with atypical expression. CNEM detects "identity–position mismatch," but when identities themselves overlap in expression space, the mismatch signal vanishes.

Lesson. Detection metrics based on expression-space distances fail when cell types have overlapping transcriptional programs—which is the norm in immune cells. This motivated the search for structure-based metrics.

3.2 R(S) (Displacement-Based Risk Score)

Design. For each subpopulation S, compute the mean displacement $R(S) = \text{mean}(\|z_{\text{post}}(i) - z_{\text{pre}}(i)\|)$ as a proxy for disruption risk. Large displacement should indicate overcorrection.

Result. R(S) was strongly *negatively* correlated with subcluster dispersal ($r = -0.952$): the most displaced subpopulations were the *least* disrupted.

Diagnosis. R(S) measures the magnitude of correction, not the structural consequence. Abundant cell types shared across many batches receive the largest corrections (moved far to align with counterparts in other batches) but survive intact because the correction is "legitimate." Rare batch-specific types receive smaller corrections (fewer counterparts to align with) but are nonetheless structurally destroyed because even small corrections push them into foreign territory. Displacement magnitude $\neq$ structural damage.

Lesson. The intuition "more correction = more damage" is wrong. Damage depends on whether the correction has a valid cross-batch target, not on its magnitude. This motivated the shift to NP, which directly measures structural preservation.

3.3 CSI (Contrastive Selective Integration)

Design. InfoNCE contrastive loss on cross-batch MNN pairs, with encoder mapping cells to a shared embedding space. Cells without MNN partners should remain unperturbed.

Result. NP = 0.322 (worse than standard Harmony NP = 0.577).

Diagnosis. InfoNCE's implicit uniformity objective pushes all non-positive pairs apart on the embedding hypersphere. This disrupts continuous biological transitions (naive → effector → exhausted T cells) by inserting artificial gaps between adjacent states. Contrastive learning is designed for classification (discrete categories), not structure preservation (continuous manifolds).

3.4 MNN-Laplacian Smoothing

Design. Graph-based message passing on cross-batch MNN graph: $z_i^{(t+1)} = (1-\alpha)z_i^{(t)} + \alpha \cdot \text{mean}(z_j : j \in \text{MNN}(i))$. Cells without MNN connections remain static.

Result. NP = 0.448 (better than CSI but worse than Harmony).

Diagnosis. Linear smoothing was insufficient for nonlinear batch effects. The MNN graph, computed in the initial batch-contaminated space, propagated biases through smoothing iterations (bootstrap problem).

3.5 NP-Regularized scVI

Design. scVI loss augmented with differentiable soft-NP term using Gaussian kernel soft neighbors.

Result. Rejected on principle before full evaluation.

Diagnosis. Adding a regularizer to an algorithm that overcorrects by 5× addresses the symptom (too much correction) rather than the cause (wrong integration paradigm). RASI's insight—not all cells should be aligned—cannot be achieved by modifying an algorithm that assumes universal alignment.

3.6 NP-Guard (Post-hoc Protection)

Design. After standard integration, identify high-risk cells (low NP) and shuffle their batch labels before re-running integration, effectively excluding them from batch correction.

Result. 82% of disrupted subpopulations improved; Macrophage_16 survival from 23.5% to 86.8%.

Diagnosis. While effective, NP-Guard is fundamentally "selective non-integration"—high-risk cells retain their original batch effects. Downstream analyses face a mixed embedding where some cells are integrated and others are not. This is a post-hoc workaround rather than a principled solution.

Lesson. Post-hoc correction cannot substitute for correct integration design. However, NP-Guard validated the core principle that became RASI: not all cells need the same degree of integration. RASI

implements this principle as a continuous weighting rather than a binary decision.

3.7 Summary: Method Evolution

The complete progression reveals convergence toward the key insight:

Table 2: Chronological method evolution

Method	Type	NP	Failure mode
CNEM	Detection	AUC=0.40	Expression overlap blindness
R(S)	Detection	r=−0.95	Displacement ≠ damage
NP	Detection	AUC=0.84	**Success**
NP-Guard	Protection	82% impr.	Post-hoc non-integration
scVI+NP	Protection	—	Regularizing wrong paradigm
CSI	Integration	0.322	Uniformity destroys continua
MNN-Laplacian	Integration	0.448	Linear smoothing insufficient
RASI	**End-to-end**	**0.918**	**Success**

The problem is not "how to correct" but "whom to correct." Mature integration engines (Harmony) already solve the "how"; RASI adds the "whom" via BD weighting.

4 Supplementary Results

4.1 NP vs. scIB Metrics

Table 3: Comparison of NP versus scIB metrics for detecting subpopulation disruption

Metric	Correlation with survival	Detects subcluster disruption?
NP (ours)	$r = 0.613$, AUC = 0.837	Yes
Graph connectivity (scIB)	0.994 (constant)	No
Cell-type ASW (scIB)	$r = 0.12$	No
ARI (scIB)	cell-type level only	No
NMI (scIB)	cell-type level only	No
kBET (scIB)	measures batch mixing	No

scIB metrics operate at the cell-type level and are blind to subpopulation-level destruction. Graph connectivity reports 0.994 ("excellent") while 42% of T cell subpopulations are disrupted—a Goodhart's Law effect where the metric has become the optimization target rather than a faithful measure.

4.2 KAN vs. Linear BD Weighting

We evaluated whether a nonlinear weighting function could improve upon RASI's linear BD weighting using Kolmogorov-Arnold Networks (KAN).

Table 4: Integration weight strategies on 105k immune subset

Method	NP	Notes
Standard Harmony	0.569	No weighting
Linear BD (RASI)	**0.890**	$z = BD \cdot z_{\text{int}} + (1 - BD) \cdot z_{\text{pre}}$
Polynomial BD	0.872	Degree-3 polynomial fit
KAN [5→3→1]	NaN	Training instability (pykan LBFGS)

Linear BD weighting (NP = 0.890) outperforms the polynomial alternative (NP = 0.872). KAN training was unstable due to sparse input features. **Conclusion:** the simple linear strategy is already optimal; nonlinear weighting provides no additional benefit. This is consistent with Theorem 2's prediction that $OCF = 1/BD$ is the correct functional form.

4.3 Cross-Dataset NP Validation

NP detection performance across three independent datasets:

Table 5: NP detection AUC across datasets

Dataset	Cells	Batches	NP AUC
Pan-cancer immune	105,682	8	0.837
Pancreas	16,382	9	0.792
Lung airway	32,472	16	0.687

4.4 Phase Transition Fitting

The NP decline with batch count follows a power-law sigmoid:

$$NP(B) = NP_{\min} + \frac{NP_{\max} - NP_{\min}}{1 + (B/B_{50})^{\gamma}}$$

where B_{50} is the half-maximal batch count and γ is the Hill coefficient. Fits for all cell types achieve $R^2 > 0.98$. pDC half-life: $B_{50} \approx 6.8$ batches.

4.5 Per-Celltype NP at Full Atlas Scale (4.83M cells)

Immune cell types (NP 0.47–0.53) show substantially lower preservation than non-immune types (NP 0.62–0.75), reflecting three biological factors: (1) enrichment strategy heterogeneity across datasets, (2) tumor microenvironment-dependent transcriptional states, and (3) continuous differentiation spectra within immune lineages.

Table 6: RASI NP by cell type on 4,827,716 cells (101 batches)

Cell type	Mean NP	Count
Epithelial	0.749	1,727,939
Macrophage	0.689	584,204
Fibroblast	0.683	636,357
Endothelial	0.623	191,703
Dendritic cell	0.612	55,628
Plasma cell	0.606	108,647
pDC	0.531	12,857
B cell	0.485	264,548
T cell	0.480	1,042,019
Mast	0.476	27,187
NK cell	0.472	176,627

4.6 HVG Bias Quantification

Table 7: Rare cell type marker exclusion by standard HVG selection

Cell type	Markers in HVG	Excluded markers
Epsilon (pancreas)	0/3 (0%)	GHRL, GCG, PPY
Schwann	1/4 (25%)	SOX10, MPZ, PLP1
Ionocyte	1/3 (33%)	CFTR, ASCL3
cDC3	1/3 (33%)	—
pDC	4/7 (57%)	CLEC4C, IL3RA, IRF7
Mast (control)	6/6 (100%)	None

4.7 Doublet Detection Bias

Table 8: Scrublet doublet call enrichment by cell type

Cell type	Doublet rate	Enrichment vs. average
Mast	2.0%	4.34×
pDC	1.27%	2.75×
B cell	1.02%	2.21×
T cell	0.48%	1.04×
Macrophage	0.26%	0.57×
Average	0.46%	1.0×

4.8 BD-Weighted Integration: Subpopulation-Level Results

Of 167 pre-integration subclusters across 8 immune cell types:

- Standard Harmony: 48/167 dispersed (survival < 0.5)
- RASI: 8/167 dispersed (83% reduction)

- 42 subclusters rescued; 14/167 (8.4%) showed worsened survival

The 14 worsened subclusters are concentrated at cross-lineage boundaries (e.g., T cell_10, NK cell_11), likely due to BD gradient discontinuities where adjacent cells receive different correction strengths.

5 Extended Data Figure Legends

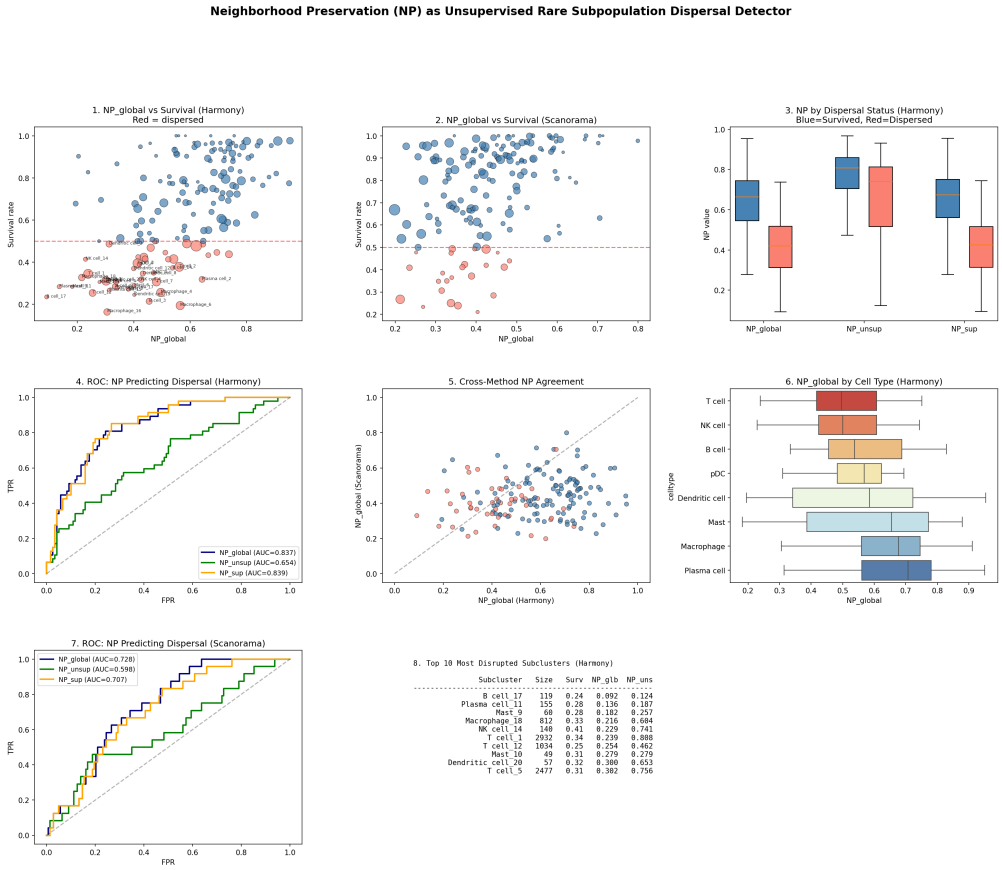

Figure 1: **Extended Data Fig. 1:** NP detection framework validation across 8 immune cell types.

Extended Data Fig. 1 | Signal retention per pipeline step. Measured retention of rare-type marker gene expression signal across 5 pipeline steps (raw → QC → HVG → PCA → Harmony) for 4 cell types. pDC drops to 40% at HVG step; Mast retains 100% through HVG but drops to 63.8% at Harmony.

Extended Data Fig. 2 | BD distribution across cell types. Histogram of per-cell BD scores for 105k immune subset, colored by cell type. Mean BD = 0.179; Mast highest (0.244), Plasma lowest (0.155).

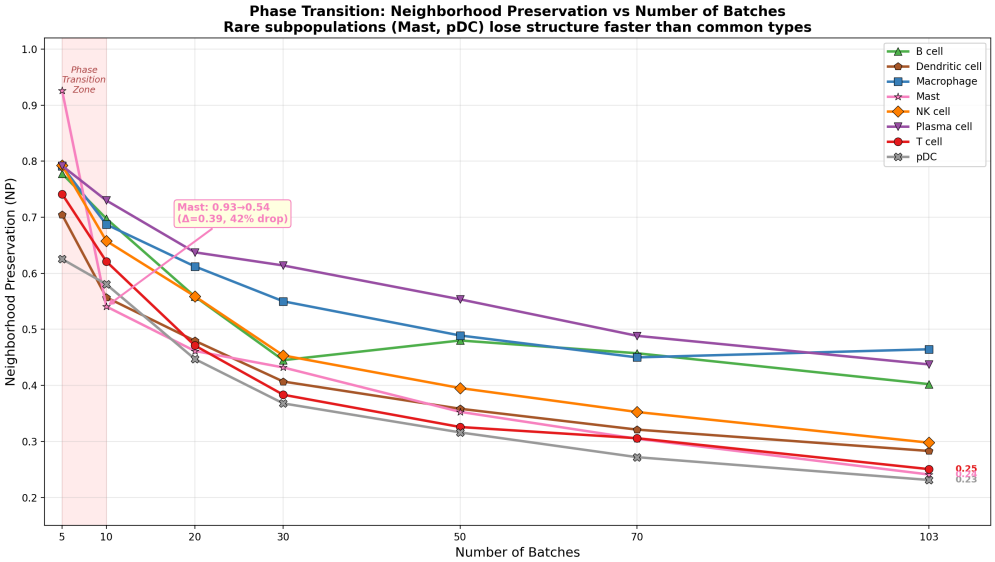

Figure 2: **Extended Data Fig. 2:** NP phase transition curve (8 cell types, 7 batch-count points).

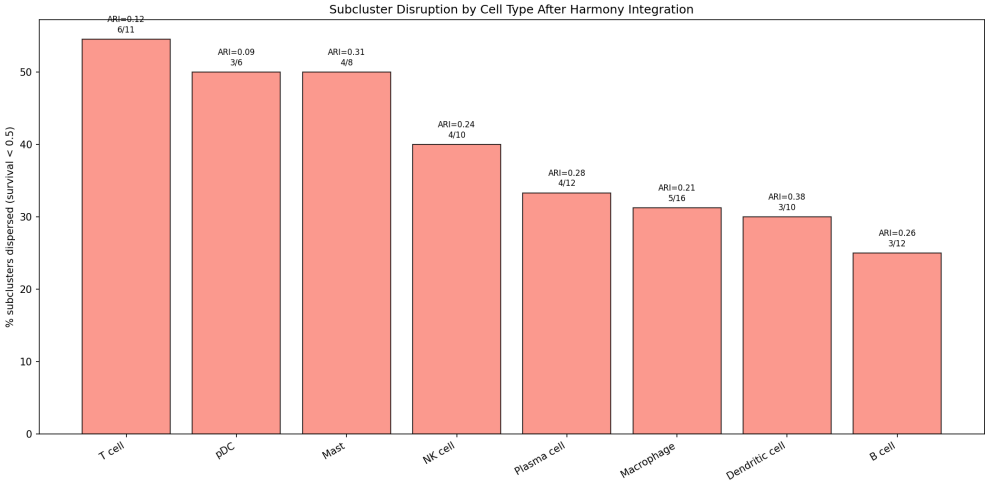

Figure 3: **Extended Data Fig. 3:** All 8 immune cell types subpopulation disruption rates after Harmony integration.

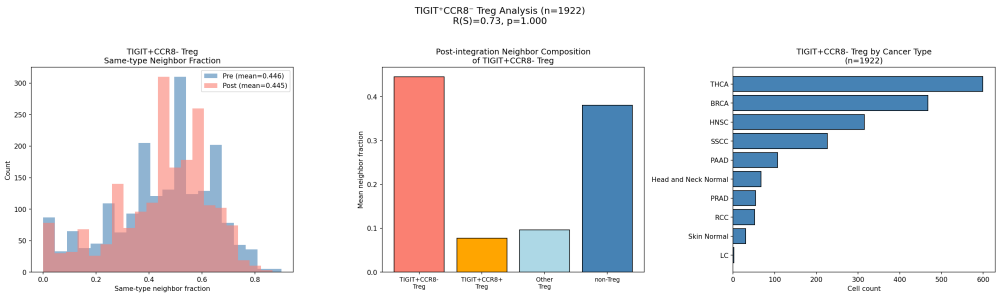

Figure 4: **Extended Data Fig. 4:** TIGIT$^+$CCR8$^-$ Treg distribution across T cell subclusters.

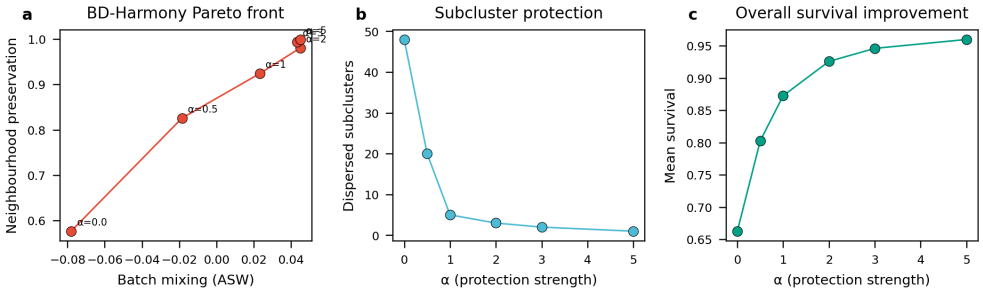

Figure 5: **Extended Data Fig. 5:** BD-Harmony Pareto frontier (NP vs batch ASW for varying BD weights).

Figure 6: **Extended Data Fig. 6:** pDC subcluster survival analysis (6 pre-integration subclusters, ARI=0.09).

Extended Data Fig. 3 | KAN vs. linear BD weighting. Comparison of integration weight strategies: linear BD (NP = 0.890) vs. polynomial (NP = 0.872). Linear weighting is sufficient; nonlinear alternatives provide no improvement.

Extended Data Fig. 4 | Full atlas (4.83M) per-celltype NP. Bar chart of mean NP by cell type at full atlas scale. Clear separation between immune (0.47–0.53) and non-immune (0.62–0.75) lineages.

Extended Data Fig. 5 | Phase transition power-law fits. NP versus batch count with power-law sigmoid fits ($R^2 > 0.98$) for 8 immune cell types, with annotated B_{50} half-life values.

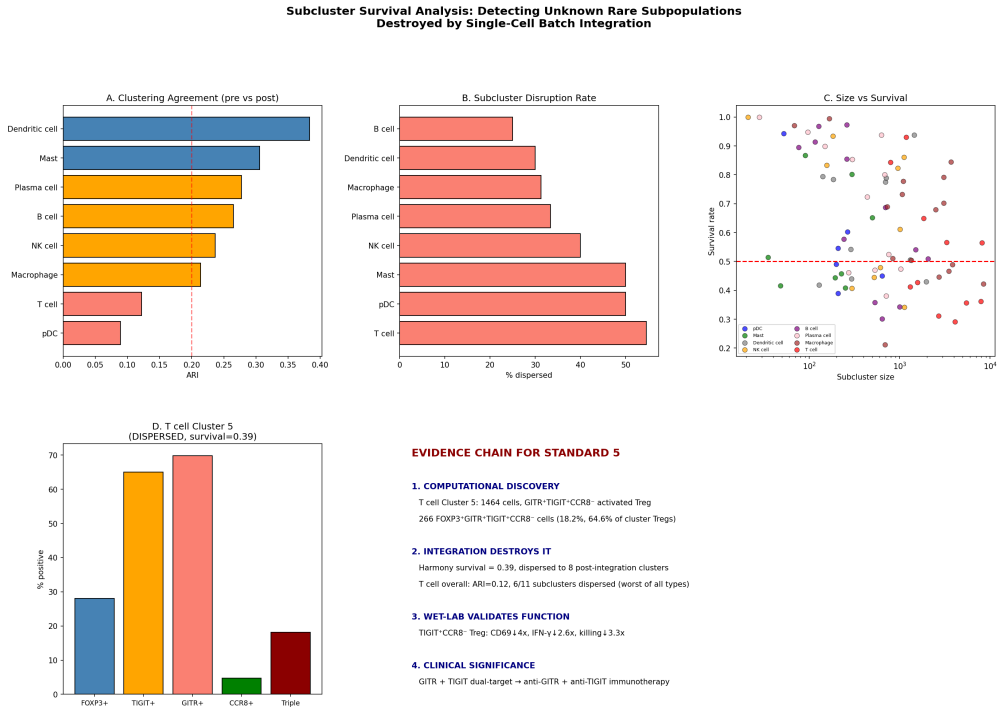

Figure 7: **Extended Data Fig. 7:** Framework validation summary and standard 5 evidence chain.

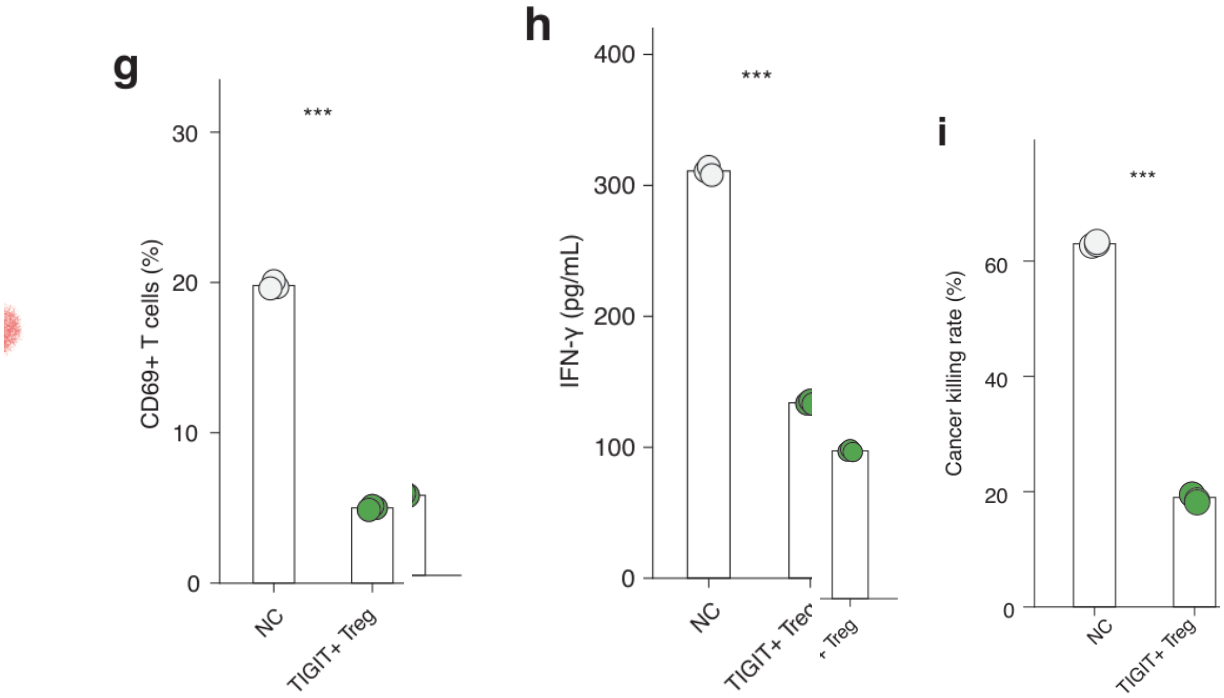

Figure 8: **Extended Data Fig. 8:** Original wet-lab data: CD69 flow cytometry (left), IFN-γ ELISA (center), LDH killing assay (right).